Vikas Kumar

IoT Embedded Systems Developer | Smart Automation Enthusiast
[vk0102103@gmail.com] · +91-9473263768 · Rajkot, Gujarat, India
GitHub · LinkedIn · Portfolio

Professional Summary

Computer Science Engineering student specializing in AI ML with hands-on experience in **IoT systems**, **embedded programming**, and **smart automation projects**. Skilled in Arduino, NodeMCU, ESP8266, sensor integration, and Firebase-based real-time monitoring systems. Developed multiple IoT and automation projects including smart irrigation and autonomous robotics solutions. Passionate about building real-world smart technologies and seeking remote internship opportunities in IoT and Embedded Systems.

Technical Skills

IoT & Embedded Systems	Arduino, NodeMCU, ESP8266, ESP32, Sensor Integration, Microcontrollers, Serial Communication, Motor Control, Smart Automation, Real-Time Monitoring, Wireless Communication
Programming Languages	Embedded C, C, C++, Python
Cloud & Tools	Firebase, Git/GitHub, VS Code, REST APIs, MongoDB
Web Technologies	HTML, CSS, JavaScript
AI & Data	Machine Learning Basics, OpenCV, Data Analysis, Weather Prediction Models

Relevant Coursework

Embedded Systems, Internet of Things (IoT), Data Structures, Computer Networks, Database Management Systems, Artificial Intelligence

Projects

Smart Irrigation System with Weather Prediction | ***Arduino, NodeMCU, Firebase, Python*** | **GitHub**

- Developed an IoT-based smart irrigation system using Arduino and NodeMCU for real-time monitoring of soil moisture, temperature, and rainfall data.
- Integrated Firebase real-time database for remote motor control, live sensor monitoring, and automated alert notifications.
- Implemented weather prediction logic in Python to optimize irrigation scheduling and reduce unnecessary water usage.
- Established reliable serial communication between sensor modules and cloud dashboard for real-time data transmission.

JalMitra – Autonomous River Cleaning Robot | ***Embedded Systems, Sensors, Automation*** | **GitHub**

- Built a solar-powered autonomous river cleaning robot using microcontrollers, sensors, and motor control systems for smart waste collection.
- Implemented obstacle detection and autonomous navigation using sensor-based embedded control logic.
- Integrated real-time water quality monitoring using turbidity and pH sensors with wireless dashboard reporting.

Smart Home Automation System | ***IoT, ESP8266, Sensor Integration*** | **GitHub**

- Developed an IoT-based home automation system for smart lighting, appliance control, and temperature monitoring.
- Implemented wireless communication and sensor-based automation for remote device management and energy-efficient operation.

Experience

PHP Developer Intern (Remote) — Dec 2025 – Feb 2026
Spacece India Foundation — *India (Remote)*

- Developed backend modules using PHP and MySQL for dynamic web applications and data management systems.
- Designed REST APIs for efficient frontend-backend communication and real-time data handling.
- Worked on database optimization, server-side logic, and system integration to improve application performance and reliability.

Education

Bachelor of Technology (B.Tech) in Computer Science Engineering – AI & ML — 2024 – Present
Marwadi University, Rajkot, Gujarat — *CGPA: 7.9 / 10.0*

Achievements

- **2nd Runner-Up** – Intellify Hackathon 2025 for IoT-based Smart Irrigation System
- **Theme Winner (Agri-Food Tech)** – Sustain-a-thon 2025 for AI & IoT Innovation Project
- Participated in multiple national-level hackathons focused on IoT, automation, and smart technology solutions

Certifications

- C++ Programming Certification
- AI Tools & Automation Certification
- Hackathon Participation and Innovation Certifications